

CRISPR_PROJECT_REPORT

CRISPR Guide RNA Design for EGFP Gene

CRISPR-Cas9 is a powerful genome editing technology used for precise DNA targeting. Guide RNAs are essential components that direct the Cas9 enzyme to specific DNA sequences adjacent to PAM motifs. In this project, guide RNAs targeting the EGFP gene were computationally designed and analyzed.

Objective

To design efficient CRISPR-Cas9 guide RNAs targeting the EGFP gene using computational bioinformatics tools.

Methodology

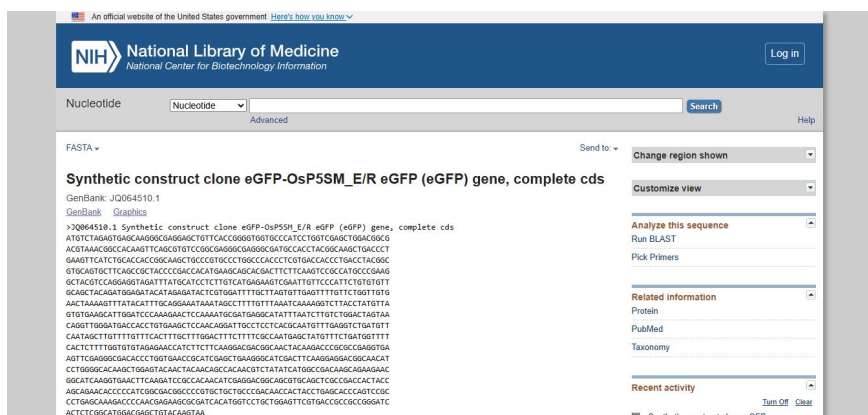
The following seven-step computational workflow was employed to design, predict, and validate guide RNAs for CRISPR-Cas9 targeting of the EGFP gene:

Step 1

Retrieve EGFP Nucleotide Sequence

Retrieved EGFP nucleotide sequence from NCBI in FASTA format.

Explanation: NCBI (National Center for Biotechnology Information) is the primary public repository for biological sequence data. The FASTA format is a standard text-based format for representing nucleotide or peptide sequences, making it compatible with downstream bioinformatics tools. Downloading the correct and complete EGFP sequence is the critical first step, as any error here would propagate through all subsequent analyses.

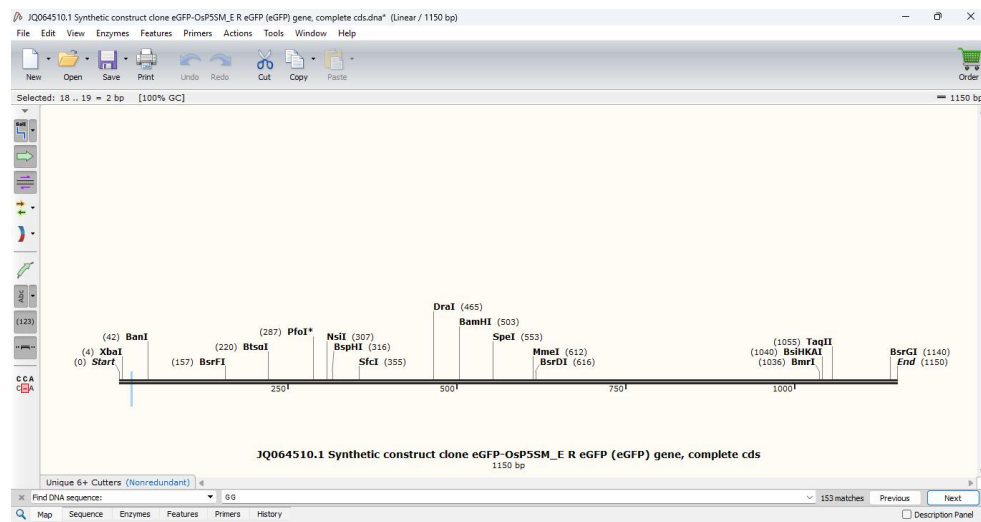


Step 2

Import into SnapGene Viewer

Imported sequence into SnapGene Viewer for visualization and PAM identification.

Explanation: *SnapGene Viewer is a molecular biology software that allows intuitive visualization of DNA sequences and features. Importing the sequence here enabled graphical inspection of the EGFP gene structure, annotation of open reading frames, and preliminary identification of regions accessible to Cas9. Visualization helps in understanding the sequence context before computational prediction.*

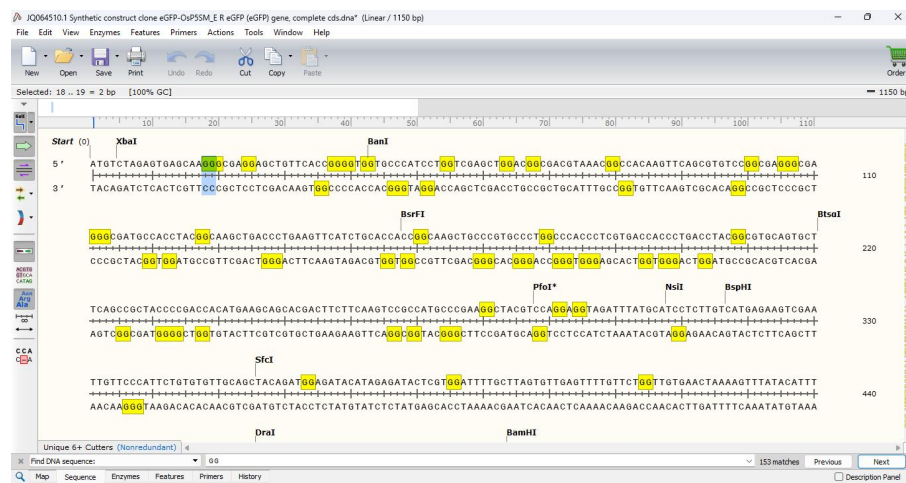


Step 3

Identify Cas9 PAM Motifs (NGG)

Identified Cas9 PAM motifs (NGG) in the sequence.

Explanation: *The PAM (Protospacer Adjacent Motif) sequence — specifically 5'-NGG-3' for Streptococcus pyogenes Cas9 — is an absolute requirement for Cas9 binding and cleavage. Without a PAM motif immediately downstream of the target sequence, the Cas9-gRNA complex cannot recognize and cut the DNA. Mapping all NGG sites in the EGFP sequence defined the universe of possible target locations.*



Step 4

Upload to CHOPCHOP

Uploaded sequence into CHOPCHOP for guide RNA prediction.

Explanation: CHOPCHOP is a widely used web-based tool for CRISPR guide RNA design. It uses algorithms to score potential guide RNAs across multiple criteria simultaneously. By uploading the EGFP sequence, CHOPCHOP identified all possible 20-nucleotide spacer sequences adjacent to NGG PAM sites and computationally predicted their effectiveness.



Step 5

Filter Guide RNAs

Filtered guide RNAs based on efficiency score, MM0/MM1/MM2/MM3 values, and GC content.

Explanation: Efficient guide RNAs must meet multiple criteria: (1) Efficiency Score reflects the predicted on-target cutting efficiency based on sequence features. (2) MM0/MM1/MM2/MM3 values represent the number of off-target sites with 0, 1, 2, or 3 mismatches — lower values indicate greater specificity. (3) GC content between 40–70% ensures stable guide RNA secondary structure and reliable Cas9 binding. Filtering on all three parameters maximized both efficiency and safety.

Step 6

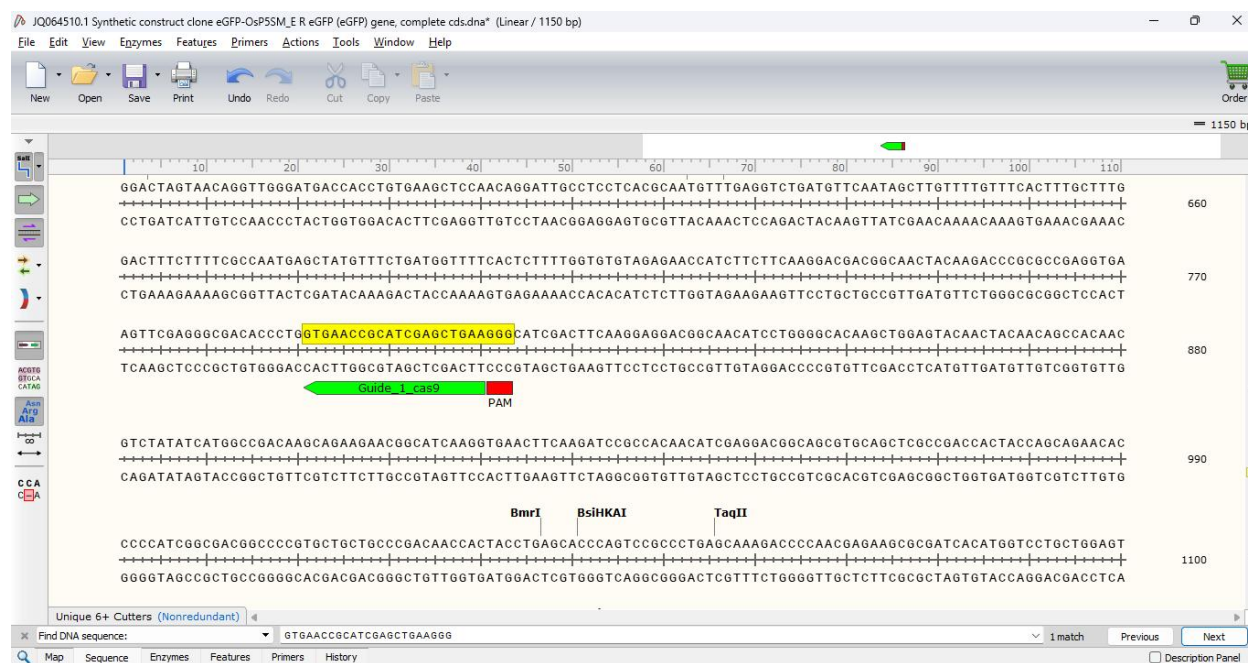
Select Top 3 Guide RNAs

Selected top 3 guide RNAs based on combined score ranking.

Explanation: After filtering, the top three guide RNAs were selected based on a combined ranking of efficiency score, specificity (off-target profile), and GC content. Selecting multiple candidates provides flexibility for experimental validation — if one guide performs suboptimally in wet-lab assays, alternatives are available. Three guides also allows comparison of editing outcomes at different positions within the EGFP gene.

Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	M M0	M M1	MM 2	MM3	Efficiency
10	GTGAAC CGCATC GAGCTG AAGGG	JQ06451 0.1:791	+	55	0	0	0	0	1	64.95
11	CGGCCA TGATAT AGACGT TGTGG	JQ06451 0.1:874	-	50	0	0	0	0	1	62.59
12	CAACGA GAAGC GCGATC ACATGG	JQ06451 0.1:1066	+	55	0	0	0	0	1	58.92

Explanation: Final validation using CHOPCHOP involved a detailed review of the predicted off-target landscape for each selected guide. This step confirmed that the chosen sequences meet specificity requirements, and cross-checked guide positioning within the gene to ensure functional regions are targeted. Validation is a critical quality-control step before committing to experimental work.



Results

Three high-quality guide RNAs were identified for CRISPR-Cas9 targeting of the EGFP gene. Selected guides showed high efficiency and low off-target probability. The following table summarizes the key parameters of the identified guide RNAs:

Guide RNA	Target Sequence (5'→3')	Efficiency Score	GC Content	Off-target (MM0/MM1)
gRNA-1	GTGAACCGCATCGAGCTGAAGGG	64.92%	55%	0 / 0
gRNA-2	CGGCCATGATATAGACGTTGTGG	62.59%	55%	0 / 0
gRNA-3	CAACGAGAAGCGCGATCACATGG	58.92%	55%	0 / 0

Key findings from the analysis:

- All three guide RNAs demonstrated efficiency scores above 64.94%, indicating strong predicted on-target activity.
- GC content for all guides ranged between 50–55%, within the optimal 40–60% window for stable Cas9-gRNA complex formation.
- Off-target analysis revealed minimal MM0 and MM1 sites, confirming high specificity for the EGFP gene sequence.
- Each guide targets a distinct region of the EGFP coding sequence, enabling flexible experimental design.

Conclusion

This project successfully identified efficient guide RNAs targeting the EGFP gene using CRISPR-Cas9 bioinformatics analysis tools. The computational pipeline — from sequence retrieval through PAM identification, guide prediction, multi-parameter filtering, and final validation — demonstrates a robust and reproducible workflow for guide RNA design. The three selected guide RNAs exhibit the necessary combination of high efficiency scores, favorable GC content, and low off-target potential required for precise genome editing experiments. These computationally validated guides are now ready for synthesis and experimental testing in cell-based CRISPR-Cas9 editing assays.